

Why breaking H₂ is hard

Bond breaking is central to chemical reactions; H₂ is the simplest test of whether DFT gets the electron rearrangement right.

1 Stretch the bond: a second configuration becomes essential

BONDED

Mostly two electrons in the bonding orbital

→

STRETCHED

Antibonding-pair configuration gains weight

→

SEPARATED

H (A)
One electron on each neutral H atom

σ_g : bonding

g = gerade (even)
Invert through midpoint:
orbital keeps its sign.

1s waves add → density builds between nuclei.

+/- show wavefunction sign (phase), not electric charge. Inversion maps each point through the bond center.

σ_u : antibonding (also σ_u^*)

u = ungerade (odd)
Invert through midpoint:
orbital flips its sign.

1s waves subtract → a node between nuclei.

READ THE SYMBOLS

σ : unchanged by rotation around the H-H axis.
 σ_g^2 : two opposite-spin electrons in σ_g .
Configuration = an orbital-occupation pattern.
g/u label symmetry, not bonding in general; an asterisk marks antibonding.

2 Watch the energy and the real-electron occupations

Energy relative to two separated H atoms

Legend: **RHF** (Restricted Hartree-Fock), **UHF** (Unrestricted Hartree-Fock), **FCI** (Full configuration interaction). **2 H** is the reference energy.

Annotations: RHF: wrong dissociation limit; Static correlation: several configurations are needed.

Interacting natural occupations

Annotation: Bonding/antibonding: (2, 0) → (1, 1).

Basis = allowed orbital building blocks; STO-3G uses one 1s (lowest atomic orbital) per H. FCI mixes all allowed configurations and is **exact in this basis**.

3 At infinite separation: correct average density, wrong pair probabilities

Restricted σ_g^2 reference

Restricted: both spins share one spatial orbital.

50% ionic weight

Ionic weight = chance both electrons are on the same atom.
Orange diagonal: charged H⁻ + H⁺; blue off-diagonal: neutral H + H.

Correlated singlet

Singlet: spins combine to zero. Measured along any shared axis, they give opposite results.

0% ionic weight

correlate

→

Cancel the ionic amplitudes

A/B: 1s orbitals on atoms A/B; (1)/(2): electron labels.

$$\sigma_g = \frac{A+B}{\sqrt{2}} \quad \sigma_u = \frac{A-B}{\sqrt{2}}$$

$$\Psi_S = \frac{1}{\sqrt{2}}(\sigma_g(1)\sigma_g(2) - \sigma_u(1)\sigma_u(2))$$

$$= \frac{1}{\sqrt{2}}(A(1)B(2) + B(1)A(2))$$

Ψ_S : the pair's spatial wave; $|\Psi_S|^2$: probability density.

Superposition: add waves first, then square to get probabilities. Same-atom terms cancel; only one-on-each-atom terms survive.

4 What this means for density-functional theory (DFT)

KOHN-SHAM (KS): ONE ORBITAL

$$\varphi_{KS} = \sqrt{\frac{n}{2}}$$

n : electron density · φ_{KS} : model orbital

Its occupation stays 2. Exact XC supplies correlation; this model electron pair is not the real state.

LOWER ENERGY, WRONG SPIN STATE

UHF lets up/down spins use different orbitals. It reaches two neutral H atoms, but the spins no longer form a pure singlet.

THE FUNCTIONAL MUST DO THE WORK

$$E_{xc}[n]$$

Exchange-correlation (XC) energy functional

A functional maps density to energy. Common approximations can miss static correlation; exact DFT need not.

H₂ exposes a central DFT challenge: the density can look right while the energy is wrong unless XC captures how electrons avoid each other.

Fixed nuclei; curves include nuclear repulsion. E_h : hartree, the atomic energy unit. Natural occupations: eigenvalues of the one-electron density matrix. Maps/formula: nonoverlapping 1s orbitals; singlet spin factor suppressed. Clouds are schematic.
Fuchs et al., JCP 122, 094116 (2005) · Burke: The ABC of DFT, §§4, 7, 11, 13 · Hehre et al., STO-3G (1969) · MIT: orbital symmetry · MIT: singlet spins